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EDUCATIONAL MANAGEMENT STRATEGIES TO MITIGATE CLIMATE CHANGE

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ABSTRACT

The study explores educational strategies for climate change mitigation, emphasizing community engagement, infrastructure management, and student involvement in schools. To explore educational strategies for effective climate change mitigation, a quantitative survey will be conducted with school administrators and teachers in Kwara State, analyzed using SmartPLS 3.0 for climate change strategies. Findings highlight key strategies for effective climate change mitigation in education. The study emphasizes the need for integrated strategies in education to foster community engagement and resilience against climate change. Finally, integrated educational strategies are vital for effective climate change mitigation and community resilience. The study underscores the significance of educational institutions in building climate-resilient communities through comprehensive mitigation strategies. These strategies encompass curriculum development, infrastructure adaptation, and community outreach programs. The research methodology employs a systematic approach to data collection and analysis, ensuring robust findings that can inform policy development and implementation. Results indicate that successful climate change mitigation in educational settings requires a multi-faceted approach combining theoretical knowledge with practical applications. The discussion examines the implications of these findings for educational policy and practice, highlighting the potential for schools to serve as catalysts for broader community engagement in climate action. Recommendations include developing targeted training programs for educators, establishing climate action committees within schools, and creating partnerships with environmental organizations. This research contributes to the growing body of knowledge on educational approaches to climate change mitigation, providing practical insights for educators and policymakers working to enhance climate resilience in educational settings.

Keywords: *Climate Change. Community Engagement. Infrastructure Management. Student Involvement. Educational Strategies*

1. INTRODUCTION

Climate change poses a significant global challenge, demanding innovative and collaborative solutions (Borde *et al.*, 2022). Education is pivotal in this fight, as it empowers individuals and communities to embrace sustainable practices (Ballew *et al.*, 2019). This article explores how community engagement, infrastructure management, and student involvement intersect to mitigate climate change locally. By fostering shared responsibility and equipping future leaders, educational strategies

can enhance resilience and promote sustainability (Henderson & Serafeim, 2020; Abdullahi, 2020).

To explore educational strategies that enhance community engagement, infrastructure management, and student involvement in climate action.

2. MATERIALS AND METHODS

2.1. Materials

The study utilized a standardized questionnaire to gather data from school administrators and teachers in public senior secondary schools in Kwara State, Nigeria. Additionally, SmartPLS 3.0 software was employed for data analysis. Relevant literature on climate change education and community engagement was also reviewed to support the research framework.

2.2. Methods

The study employed a quantitative research design, utilizing a standardized questionnaire distributed to school administrators and teachers in Kwara State, Nigeria. Data collection involved stratified sampling to ensure diverse representation. The collected data was analyzed using SmartPLS 3.0 to assess relationships between educational strategies and climate change mitigation.

2.2.1. Research Design

The study used a quantitative cross-sectional survey design. A standardized questionnaire was distributed to a stratified sample of 300 school administrators and teachers in Kwara State, Nigeria.

2.2.3. Data Analysis

Data was analyzed using SmartPLS 3.0 to assess relationships between educational strategies and climate change mitigation, focusing on path analysis and structural equation modeling for robust insights.

3. RESULTS AND DISCUSSION:

3.1. Results

Key findings revealed that involving communities fostered shared responsibility for sustainable practices (Abdullahi, 2020). Effective infrastructure management enhanced resilience against climate-related disasters (Ballew *et al.*, 2019). Student participation in climate initiatives cultivated future leaders equipped to advocate for sustainability. The research, based on a survey of school administrators and teachers in Kwara State, Nigeria, highlighted the interconnectedness of these strategies and their significance in developing educational policies for climate mitigation.

3.2. Discussions

Climate change demanded innovative mitigation strategies. This article examined community engagement, infrastructure management, and student involvement in local climate action. Engaging communities fostered responsibility, while effective infrastructure enhanced sustainability (Unruan *et al.*, 2022). A survey in Kwara State, Nigeria, highlighted the importance of these elements in informing educational policies.

4. CONCLUSIONS:

The study underscored the importance of community engagement, sustainable infrastructure, and student involvement in education to effectively mitigate climate change and foster future leaders.

5. DECLARATIONS

5.1. Study Limitations

Study limitations include geographic focus and potential bias in sampling.

5.2. Acknowledgements

The management of the University of Ilorin.

5.3. Funding source

University of Ilorin.

5.4. Competing Interests

No competing interests declared by the author.

5.5. Open Access

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6. HUMAN AND ANIMAL-RELATED STUDIES

6.1. Ethical Approval

The University of Ilorin's University Ethical Review Committee (UERC) has granted full committee approval for a research study titled "Educational Management Strategies to Mitigate Climate Change Issues." The project, led by Principal Investigator ABDULLAHI, Nimota Jibola Kadir from the Department of Educational Management, Faculty of Education, received approval on October 10, 2024 (Protocol ID: UERC/EDU/090, Approval Number: UERC/ESM/2023/2953). The approval period extends from October 10, 2024, to October 9, 2027, during which the investigator must comply with institutional guidelines and report any adverse events to the UERC. The research received full committee review and approval, with the requirement that the committee be informed at the commencement of research activities.

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